

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Designed and shipped to production an agentic AI solution over 67 BigQuery tables with **multi-agent orchestration** (Context Researcher → SQL Drafter → Validator → Devil's Advocate), 8 deterministic rules, and a 23-case golden-query evaluation suite; translated ambiguous enterprise business problems into an AI system architecture stakeholders adopted. Approved by Sr. Director and product leadership; currently the only AI skill in active use by business teams on Data Ventures.
- Built and shipped an autonomous Jira ticket-resolution agent with **RAG pipeline architecture**: 6-gate triage, MCP-driven context gathering, hybrid FAISS + BM25 retrieval over 11,000 historical SQL queries, and self-healing execution; closed 400+ requests across 8 categories over 15 months and cut routine turnaround from 30–60 minutes to under 10. Designed it as reusable patterns and orchestration components, not one-offs.
- Architected a registry-governed semantic layer that embeds **AI governance and policy enforcement** into the design: strict-JSON intent with schema validation (the model never writes SQL), deterministic validation/canonicalization, and a human review gate — responsible-AI guardrails and reproducibility on sensitive, governed data.
- Built a hybrid orchestrator that routes enterprise questions across two backends, reasons about disagreements, and refines definitions over time; maintained reusable solution patterns and reference designs that let Product self-serve requests previously filed as tickets.
- Translated business and operational requirements into AI system designs for technical and non-technical stakeholders, led working sessions, and drove adoption; delivered 5 Power BI dashboards ranked top 1.4% of Walmart's 207,000 dashboards (1,064 views, 22 users); Making a Difference Award. Daily LangChain + Claude (Opus/Sonnet) + Gemini builder.

FTI Consulting — UChicago MSADS Capstone

Chicago, IL · Jan – Aug 2024

- Delivered a customer-facing **RAG-to-proposal** solution for an enterprise consulting client, deployed on AWS EC2; scoped the use case, produced the proposal/POC, and owned delivery end-to-end; cut proposal time ~30%; **Best in Show — UChicago MSADS Capstone**.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) generative-AI solution for a 500+ participant experiment; controlled A/B tests on prompt-driven behavior persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + LangChain-orchestrated RAG over historical queries, BigQuery execution, auto-visualization): a deployable reference design for NL-to-data. Live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Agentic AI solution design: multi-agent orchestration patterns, RAG pipeline architecture, LLM-powered workflows, context engineering, prompt-engineering patterns, model/multi-provider selection, reusable reference designs & solution patterns, end-to-end AI system architecture

AI governance: inference governance, policy enforcement / guardrails, responsible-AI, schema validation & data contracts, human-in-the-loop review, evaluation (golden sets, regression suites)

Frameworks / stack: LangChain, RAG (embeddings + FAISS + BM25 + RRF), MCP server design, FastAPI, Cloud Run, GCP / AWS; *NVIDIA AI Enterprise (NIM/NeMo/Triton) and CrewAI/AutoGen/Semantic Kernel are ramp areas, not claimed*

Client-facing: enterprise business-problem-to-architecture translation, customer-facing proposals & POC designs, stakeholder workshops, executive communication

Data / dev: BigQuery, dbt, SQL, Python, PySpark, Airflow, Databricks; CI/CD (pytest), Git; Claude (Opus/Sonnet/Haiku), Gemini (2.5 via Vertex AI), OpenAI; Cursor, Claude Code

Viz: Power BI (DAX, XMLA TMSL), Tableau, Streamlit

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL